

Sahil Minhas

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EDUCATION

Wilfrid Laurier University
Honours Bachelor of Computer Science

Expected Dec 2026
Waterloo, Ontario

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL, JavaScript, HTML, CSS, TypeScript

Frameworks/Tools: React, Next.js, Node.js, MySQL, Swift, Supabase, MongoDB, Tailwind, Postman, Figma, Git, GitHub, OpenCV, PyTorch, TensorFlow

PROJECTS

SpendMe – Personal Finance App & Website

[Live Site](#) — [Youtube](#)

- Co-developed a native iOS personal-finance app using **Swift, SwiftUI, and SwiftData**, building dashboards, budgeting, and savings-goal workflows with persistent local storage and **Supabase** authentication and cloud synchronization.
- Implemented **calendar-based savings calculations in Swift** to derive daily, weekly, monthly, and yearly contributions from goal deadlines, integrating accrued allocations and remaining budget reservations into available-to-spend calculations.
- Built the public landing page using **React, TypeScript, and Next.js** to showcase product features and collect early-access registrations, attracting **50+ sign-ups**; created a promotional video ad to drive awareness of the app and website.

BudgetBridge – Personal Expense Tracker

[GitHub](#)

- Developed a **full-stack personal expense tracker** using **JavaScript, Node.js, and MySQL**, implementing account authentication, transaction and category CRUD operations, financial dashboards, search, filtering, and date-range reporting.
- Designed a **MySQL relational database with 5 interconnected tables** and integrated parameterized SQL queries, user ownership checks, and persistent session management to securely isolate financial data across authenticated users.
- Built and validated REST-style API workflows for authentication, transactions, categories, reports, and user profiles, achieving **12/12 passing automated tests** across CRUD, security, filtering, and cross-user isolation.

Distributed Bulletin Board System

[GitHub](#)

- Developed a **multithreaded client-server system** in **Java** using **TCP sockets**, supporting **10+ concurrent clients**.
- Designed a custom application-layer protocol with structured OK/ERROR responses, enabling **robust command parsing, validation, and error handling** while ensuring reliable and consistent communication between clients and server.
- Implemented **thread-safe synchronization** using **synchronized methods**, ensuring **atomic operations** for POST, CLEAR, and SHAKE while maintaining consistent shared state and preventing race conditions under concurrent access.

N-Puzzle Solver – AI Search Algorithm

[GitHub](#)

- Implemented a **Python** based N-Puzzle solver utilizing the **A* search algorithm** with Manhattan Distance and Misplaced Tile heuristics to efficiently explore state spaces, optimize search depth, and minimize overall computation time.
- Applied graph traversal, priority queues, and heuristic optimization techniques to develop intelligent search strategies that **reduced redundant node expansions, enhanced runtime efficiency, and improved scalability** across puzzles.
- Designed a modular architecture for state representation, node expansion, and dynamic cost evaluation that achieved a **92% reduction in nodes explored** and a **15% lower branching factor** using Linear Conflict compared to Misplaced Tile.

Image Processing Toolkit

[GitHub](#)

- Created in **Python** using interpolation, intensity transformations, and spatial filtering techniques implemented with **NumPy** based matrix operations and **OpenCV APIs** to compare performance and algorithmic efficiency across **20+ test images**.
- Used **NumPy** for optimized matrix operations and data handling, reducing computation time on pixel-based transformations.

WORK EXPERIENCE

IT Technician

S.M I.T Services

May 2024 – Aug 2025

Brampton, Ontario

- Diagnosed and resolved issues on **50+ PCs and laptops** with systematic troubleshooting methods and tools such as **Windows utilities, Linux command line** diagnostics, and hardware testing equipment to deliver effective and fast solutions.
- Cultivated strong client relationships through personalized technical support and communication, achieving a **70% increase** in repeat business and maintaining high customer satisfaction through clear documentation and consistent follow-ups.

Academic Tutor

Pathways 2 Excellence Tutoring

June 2018 – Aug 2025

Brampton, Ontario

- Provided individualized tutoring to **200+ students** across mathematics, science, and computer science subjects, increasing average academic performance by **30%** through customized lesson planning and improving final academic grades by **20%**.